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HCL TECHNOLOGIES

ENTERPRISE AI OPERATING SYSTEM

Complete Solution Proposal — RTX Corporation

$268B	$2.2B	18 Months
RTX Backlog	5-Year Return	Payback Period

Collins Aerospace	Pratt & Whitney	Raytheon
Avionics · Aviation Systems · Mission Systems	GTF Engines · Military Propulsion · MRO	Missiles · Defense · Sensors · C2 Systems

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DEAL AT A GLANCE

RTX Corporation × HCL Technologies | Enterprise AI Operating System

PROBLEM

835 aircraft grounded — 32% of the entire GTF-powered fleet

$6–7B in GTF engine quality crisis (contaminated powder metal)

250–300 day MRO turnaround vs. 60-day customer expectation

500+ fragmented applications across 3 business segments

14,000+ suppliers — zero AI-powered risk intelligence today

$200M ITAR penalty for 750+ violations — August 2024

62% of IT budget consumed by legacy maintenance, not innovation

SAP ECC mainstream support ends December 2027 — clock is ticking

HCL SOLUTION

Enterprise AI OS — 7-layer intelligence fabric across all 3 segments

AI Force (patented) — GenAI + Agentic AI, on-prem ITAR-native

Base90 — delivers 90% of OneRTX PST S/4HANA out-of-box

iMRO/4 — GTF turnaround: 250 days → 175 days (30% reduction)

6 AI agents: Manufacturing, Supply Chain, Finance, Defense, HR & Safety, IT Ops — all with hard human override governance

7 ITAR-compliant US delivery centres + Butler Aerospace clearances

SAP GSSP + Azure Expert MSP + AWS Premier Partner — one firm

INVESTMENT

$300–500M | 5-Year Phased Engagement

Phase 1 (Year 1): $60–80M

SAP S/4HANA AI integration + GTF quality AI at 3–5 lighthouse factories

Phase 2 (Years 2–3): $120–180M

Enterprise AI OS across all 3 segments + agentic supply chain + defense AI

Phase 3 (Years 4–5): $120–180M

Connected Factory AI + digital twins + full autonomous operations

RETURN

$2.2B Baseline | 18-Month Payback

3.9x ROI at baseline | Up to $3.1B optimistic scenario

$1.25B+ IT budget freed (62% → 38% maintenance)

$250M+ defense contract win rate improvement

$200M+ manufacturing downtime reduction (35%)

$150M+ supply chain risk avoidance

$125M+ MRO aftermarket pricing optimisation

$100M+ GTF engine blade scrap reduction (30%)

The only firm combining ITAR-cleared US delivery centres + patented AI platform (AI Force) + SAP GSSP with A&D templates (Base90 / iMRO/4) + #1 A&D engineering heritage by Zinnov

1 Executive Summary

835 aircraft grounded. $6–7B in GTF engine costs. 250–300 day inspection backlogs. $200M in ITAR penalties. RTX needs an Enterprise AI Operating System — not another pilot programme, not another point solution. HCL Technologies is the only partner positioned to deliver it.

HCL Technologies proposes a comprehensive Enterprise AI Operating System (EOS) for RTX Corporation — a transformative, phased engagement that addresses the full spectrum of RTX's digital fragmentation, technical debt, and AI maturity challenges across Collins Aerospace, Pratt & Whitney, and Raytheon Defense. Our solution is built on four non-negotiable pillars:

(1) An Agentic AI Operating Layer powered by HCL AI Force — unifying 500+ fragmented applications under six domain-specific AI agents with hard human override governance.

(2) An SAP S/4HANA consolidation roadmap using HCL's Base90 A&D templates — delivering 90% of OneRTX PST requirements out-of-box, ahead of SAP ECC's December 2027 end-of-support.

(3) A federated OT/IT convergence architecture using HCL iMRO/4 — connecting 250,000+ instrument tags to enterprise AI and targeting a 30%+ reduction in P&W's 250-300 day GTF inspection turnaround.

(4) A structured Centre of Excellence (CoE) that transfers full AI ownership to RTX by Month 36 — building self-sufficiency, not dependency.

1.1 — Transformation at a Glance

The table below captures RTX's current state versus HCL's committed EOS targets at the 36-month milestone. Every figure is tied to a specific KPI in the Commercial Model (Section 6) with outcome-based financial incentives — HCL's fee is linked to RTX achieving these numbers.

Dimension	Current State	HCL EOS Target (36 Months)
Applications	500+ fragmented standalone apps	**120–180 rationalised core systems**
IT Budget on Maintenance	62% consumed by legacy maintenance	**38% — freeing $250M+ for innovation**
AI Maturity	Level 2 — Experimenting (disconnected POCs)	**Level 4 — Operationalised enterprise AI**
Financial Close	10+ business days	**6 business days**
OT/IT Integration	250,000+ tags disconnected from AI	**Real-time AI-connected OT fabric**
ERP Landscape	Multiple SAP ECC + Oracle EBS instances	**Unified SAP S/4HANA with AI layer**
L1 Ticket Deflection	0% AI-assisted	**60% AI deflection via agentic IT ops**
CoE Ownership	No internal AI capability	**Fully RTX-owned CoE by Month 36**

2 Solution Architecture — Enterprise AI Operating System

RTX's 250,000 OT instrument tags stream through the OT/IT Convergence layer (L2) into a unified data lakehouse (L3), where HCL AI Force trains domain-specific models (L4) powering six agentic AI workers (L5) — each governed by RTX's Tier 1/2/3 framework with hard human override triggers. Every AI output passes through HCL's built-in GRC module before reaching any production system. No layer is optional; the intelligence compounds across all seven.

2.1 — Seven-Layer EOS Architecture

The architecture overlays intelligence across RTX's full technology estate — from GTF engine test cells at Pratt & Whitney, to missile assembly at Raytheon's Redstone facility, to Collins avionics plants — without disrupting live production. HCL proprietary IP appears at every layer, ensuring RTX is not dependent on third-party platform roadmaps for its core AI capability.

Layer	Name	What It Does	HCL Proprietary IP	Key Third-Party OEMs
L1	Infrastructure & Edge	On-prem, hybrid cloud, and edge compute across 500+ RTX facilities. ITAR-compliant US/allied-nation data residency enforced at the infrastructure level. Sub-10ms latency for real-time quality AI at engine manufacturing cells.	**HCL AI Force (on-premises deployment mode) — ITAR-native, no public cloud egress for classified model weights or inference outputs. Runs within FedRAMP High / DoD IL5 / air-gapped environments from Day 1.**	Microsoft Azure Government (FedRAMP High / DoD IL5), AWS GovCloud (US), Dell EMC PowerEdge edge servers
L2	OT/IT Convergence	Connects 250,000+ instrument tags, DCS/MES, CNC machining centres, GTF engine test cells, missile assembly lines, and additive manufacturing cells to the enterprise data fabric via OPC-UA and MQTT. Zero production system disruption.	**HCL iMRO/4 — intelligent MRO deployed at HAESL (Feb 2026) and Jet Aviation; covers P&W engine inspection, work-order AI, and MRO throughput — directly targeting the 250–300 day GTF inspection backlog.**	Siemens Industrial Edge + Opcenter MES, AVEVA PI System (OSIsoft), Rockwell FactoryTalk, PTC ThingWorx IIoT
L3	Data Platform	Unified data lakehouse ingesting operational, transactional, PLM, HR, and supply chain data from all three RTX segments. Master data governance, ITAR-aware data mesh with automated cross-border controls, and real-time + batch pipelines.	**HCL AI Force.Data — data engineering module with RAG Builder for contextual AI pipelines; OpenSearch vector embeddings for RTX engineering and contract document retrieval at scale.**	Databricks Delta Lake + MLflow, SAP BTP Data Intelligence, Azure Purview
L4	AI / ML Core	Federated model training, MLOps pipeline, model registry, drift detection, bias monitoring, and Tier 1/2/3 risk governance. Supports both commercial cloud and air-gapped classified model environments without architectural compromise.	**HCL AI Force (patented GenAI + Agentic AI platform) — model-agnostic (Azure OpenAI, Claude, Llama 3, Granite). On-premises deployment for ITAR model weights. Built-in GRC module: PII, toxicity, bias, prompt-injection scanners. HANASmart automates**	SAP AI Core + Generative AI Hub, NVIDIA AI Enterprise + DGX Systems (on-prem GPU), DataRobot MLOps + AI Governance

			ABAP-to-S/4HANA migration in days.	
L5	**Agentic Orchestration**	Multi-agent AI layer orchestrating six domain-specific agents: Manufacturing, Supply Chain, Finance, Defense Programs, HR & Safety, and IT Ops. Human-in-the-loop enforced for all Tier 1 decisions. Agent Studio enables RTX-specific custom agent development.	**HCL AI Force Agent Studio — custom AI agent builder with reusable component library. HCL AI Force.Ops — IT operations automation powering the IT Ops Agent (60% L1 ticket deflection target across 50,000+ monthly tickets).**	SAP Joule AI Copilot, Microsoft Copilot Studio + Azure AI Foundry, LangChain / AutoGen orchestration
L6	**Integration**	API gateway, event-driven middleware, and pre-built connectors for SAP ECC, Oracle EBS, Salesforce, Workday, ServiceNow, Siemens Teamcenter, PTC Windchill, and SIPR/JWICS boundary adapters for classified network integration.	**HCL iSCM — intelligent Supply Chain Management pre-integrated with SAP Ariba and IBP for RTX's 14,000+ supplier network. HANASmart integration suite for legacy ECC-to-S/4HANA connectivity during OneRTX PST build phase.**	SAP BTP Integration Suite, MuleSoft Anypoint, Apache Kafka (event streaming), IBM App Connect
L7	**Experience & Insight**	Role-based AI cockpits for C-suite, plant managers, engineers, and operators. AR/VR maintenance overlays for P&W MRO technicians. Voice-commanded SAP queries. Real-time dashboards linking OT sensor data to board-level KPIs.	**HCL AI Force.Software — engineering co-pilot for RTX development teams: 30% faster SDLC, 45% accelerated testing. Supports Collins Code Copilot and test case automation initiatives already in progress.**	SAP Analytics Cloud, Microsoft Power BI, PTC Vuforia AR, Microsoft HoloLens 2 (DoD-approved), ServiceNow AI Ops

2.2 — Agentic AI Domain Agents

Six domain-specific AI agents operate with bounded autonomy. Every agent is powered by HCL proprietary IP — AI Force, iMRO/4, iSCM, iPEO, or the AI Force GRC Module — ensuring the intelligence layer is native HCL, not a configuration of third-party tools. Human override triggers are hard-coded, not configurable, for all Tier 1 safety-critical decisions.

AI Agent	Domain	Key Capabilities	HCL IP Powering This Agent	Human Override & Governance
Manufacturing Agent	Pratt & Whitney / Collins / Raytheon Plants	AI computer vision for GTF blade inspection (micron-level); missile assembly QC; additive manufacturing monitoring; 35% downtime reduction target across 500+ facilities.	**HCL AI Force (CV inference) + iMRO/4 (MRO work-order AI) deployed at HAESL Feb 2026 — directly replicable for P&W GTF inspection backlog reduction from 250–300 days.**	Tier 1 — Full human approval for any process control action; kill-switch + real-time monitoring + rollback plan mandatory.
Supply Chain Agent	Global 14,000+ Supplier Network	Multi-tier supplier risk monitoring; rare material (Ti/Co/Re) sourcing intelligence; ITAR compliance automation; engine	**HCL iSCM (Intelligent Supply Chain Management) + AI Force.Data — RAG-enabled supplier document retrieval. Maps to HCL's DARPA RSDN-**	Tier 2 — Human approval for sole-source switches; Supply Chain Director sign-off for rare material re-sourcing.

		aftermarket demand prediction; tariff exposure mitigation ($850M 2025 exposure).	**equivalent supply chain resilience architecture.**	
Finance Agent	RTX Finance & Defense Accounting	Financial close from 10 to 6 business days; AI-powered EVMS reporting; CAS compliance monitoring; EAC prediction for $107B defense backlog; automated SOX audit trail.	**HCL AI Force for SAP — natively integrated with S/4HANA EVMS and CAS modules. HANASmart automates migration of legacy ECC financial records to S/4HANA for OneRTX PST.**	Tier 2 — SOX-compatible audit trail on all outputs; CFO/Controller approval gate for financial close automation.
Defense Program Agent	Raytheon Defense Contracts	AI-augmented contract pricing (cost-plus / FFP); win rate improvement; government profitability prediction; program EAC management. All outputs remain within FedRAMP High boundary.	**HCL AI Force GRC Module — classified data governance with input/output scanning, PII detection, prompt-injection prevention. On-premises for SIPR-adjacent data — no third-party platform dependency.**	Tier 2 — Program Director approval for pricing decisions; ITAR-cleared personnel only; classified boundary enforced.
HR & Safety Agent	185,000+ Employee Workforce	Security clearance pipeline optimisation (8–24 month backlog reduction); engineering talent retention prediction; workforce planning for defense ramp-ups; safety compliance monitoring across 500+ facilities.	**HCL AI Force.Data + HCL iPEO (Intelligent Production, Engineering & Operations) — workforce-to-production alignment integrating Workday, SuccessFactors, and clearance management databases.**	Tier 3 — HR Manager approval for talent actions; quarterly model accuracy review; basic logging + feedback loop.
IT Ops Agent	500+ Application Landscape	60% L1 ticket deflection (50,000+ monthly tickets); classified/unclassified boundary management; AI threat detection; application rationalisation from 500+ to 120–180 core systems.	**HCL AI Force.Ops — 25% faster IT issue resolution across 25+ enterprise clients. RAG Builder captures institutional knowledge from RTX's legacy application documentation ecosystem.**	Tier 1 — Security team override for classified boundary actions; zero-trust enforcement on all cross-boundary automation.

3 OEM Recommendations — Technology Stack

HCL Technologies leads with proprietary IP — AI Force, Base90, iMRO/4, HANASmart, iSCM, and the AI Force GRC Module — before layering in the best third-party platforms. Every OEM is selected because it has been evaluated against RTX's specific regulatory, operational, and AI maturity requirements, and because it extends — not duplicates — RTX's existing technology investments.

3.0 — HCL Proprietary Layer (Leads the Stack)

Unlike competitors who assemble third-party tools, HCL brings six proprietary assets that form the load-bearing intelligence core of the RTX EOS. No competitor in this bid — Accenture, IBM, TCS, or Infosys — can offer this combination of patented AI platform, SAP GSSP credentials, ITAR-cleared delivery centres, and aerospace-specific IP proven in production at HAESL and Jet Aviation.

HCL Proprietary Asset	What It Is & How It Works	Why It Matters for RTX Specifically
AI Force	Patented GenAI + Agentic AI platform. Model-agnostic (Azure OpenAI, Claude, Llama 3, Granite). On-premises deployment for ITAR-classified inference. Built-in GRC: PII, toxicity, bias, prompt-injection scanners. Agent Studio for RTX-specific custom agents. 4 modules: AI Force.Software, AI Force.Ops, AI Force.Data, AI Force for SAP.	**Addresses RTX's $200M ITAR penalty history — compliance-by-design from Day 1. Live with 25+ enterprise clients. Deployable within AWS GovCloud / Azure Gov / air-gapped environments. Satisfies NDAA Section 1513 AI/ML cybersecurity mandate ahead of June 2026 deadline.**
Base90	Preconfigured SAP S/4HANA A&D templates covering 90% of RTX requirements out-of-box. Covers Plan-to-Sustain: manufacturing, logistics execution, and SAP GTS (Global Trade Services) for ITAR/EAR export compliance within S/4HANA.	**Directly accelerates OneRTX PST build phase — currently in build/support phase targeting ECC end-of-support December 2027. Reduces custom development risk by 60%+ and compresses the build timeline by an estimated 8–12 months.**
iMRO/4	Intelligent MRO solution built as an SAP add-on covering 6 core process areas: work-order management, inspection scheduling, spare parts optimisation, supplier MRO collaboration, technician task management, and MRO cost tracking.	**Selected by HAESL (February 2026) and deployed at Jet Aviation. Directly replicable for P&W GTF engine inspection — target: reduce 250–300 day turnaround by 30%+, saving $1–1.5B annually at P&W MRO scale.**
HANASmart + GenAI	Automates migration of custom ABAP code from SAP ECC to S/4HANA in days instead of months. Co-created with SAP at HCL's AI & Cloud Native Lab in Munich (launched December 2024). Eliminates highest-risk element of any S/4HANA migration.	**RTX runs multiple large SAP ECC instances across P&W and Collins. HANASmart removes the manual ABAP re-coding bottleneck — directly addressing the costliest and most time-consuming element of OneRTX PST's build phase.**
iSCM	AI-powered Intelligent Supply Chain Management pre-integrated with SAP Ariba and IBP. Covers demand sensing, supplier risk scoring, multi-tier visibility, rare material sourcing intelligence, and ITAR-compliant supplier data handling across 14,000+ suppliers.	**Addresses RTX's $850M 2025 tariff exposure and 14,000+ supplier network complexity. Provides AI-driven early warning for sole-source bottlenecks — structural castings, isothermal forgings, microelectronics — that are the primary cause of aircraft groundings.**

AI Force GRC Module	Built-in governance: PII, toxicity, bias, and prompt-injection scanning on all AI inputs/outputs. SSL/TLS 1.3, LDAP integration, classified data boundary enforcement. Four deployment modes: cloud, hybrid, on-premises, and edge.	**Replaces need for Palantir Foundry for classified data governance. Runs within FedRAMP High / DoD IL5 / air-gapped environments. Directly addresses RTX's $200M ITAR penalty history — zero data egress from classified environments, provable audit trail on every AI inference.**

3.1 — Core ERP & AI Platform (SAP Backbone)

HCL Technologies is a SAP Global Strategic Services Partner (GSSP) — the highest partnership tier globally — with an estimated $2.5–3B in SAP-related revenue and 15,000–20,000 SAP consultants. HCL co-created an AI & Cloud Native Lab for SAP Business AI in Munich in December 2024. For RTX's OneRTX PST programme, no competitor can match HCL's combination of Base90 A&D templates, HANASmart migration automation, and AI Force for SAP.

OEM	Product / Module	Role in RTX EOS	Why Selected for RTX
SAP	**S/4HANA Cloud Private Edition**	ERP consolidation backbone — replaces multiple SAP ECC and Oracle EBS instances across all 3 segments. Central financial, manufacturing, supply chain, and HR platform.	HCL Technologies is a SAP GSSP — the highest tier globally — with 15,000–20,000 SAP consultants. HCL's Base90 A&D templates deliver 90% of OneRTX PST requirements out-of-box, reducing build risk for the December 2027 ECC deadline.
SAP	**SAP BTP (AI Core + Integration Suite + Data Intelligence)**	AI model hosting, MLOps, API integration hub, and data pipeline orchestration connecting all 500+ RTX systems. SAP Generative AI Hub for multi-LLM access with governance.	Native SAP integration eliminates middleware risk. AI Core hosts RTX-specific aerospace models. HCL's AI & Cloud Native Lab (Munich, SAP co-created, December 2024) built specifically to deliver this architecture at enterprise scale.
SAP	**SAP Joule AI Copilot + Joule Studio**	Natural language AI interface — SAP queries via voice, GTF work orders via conversational AI, EVMS reporting via chat. Joule Studio enables no-code agent building for multi-step autonomous processes.	First AI copilot natively embedded across S/4HANA, BTP, SuccessFactors, and Ariba — no custom build required. Extensible with RTX-specific aerospace context via HCL's AI Force for SAP integration layer.
SAP	**SAP Ariba + SAP IBP**	Procurement AI and integrated business planning for 14,000+ suppliers. Demand-supply balancing for GTF MRO aftermarket. Rare material (cobalt, tantalum, rhenium) sourcing intelligence.	Largest procurement network globally. HCL iSCM pre-integrated with Ariba and IBP, adding agentic supply chain intelligence without duplicate licensing or integration cost.

3.2 — Cloud & Infrastructure (ITAR-Compliant)

RTX's cloud strategy is anchored to AWS (December 2025 Strategic Collaboration) and Microsoft Azure (existing enterprise collaboration). HCL holds the highest partnership tier with both — Azure Expert MSP with all 21 Microsoft specialisations, and AWS Premier Consulting Partner — maximising RTX's existing investments rather than duplicating them.

OEM	Product / Module	Role in RTX EOS	Why Selected for RTX
Microsoft	**Azure Government (FedRAMP High / DoD IL5)**	Primary cloud for US classified-adjacent workloads. SAP S/4HANA on Azure for OneRTX PST hosting. AI/ML workloads on Azure AI Studio. M365 enterprise	Only hyperscaler with DoD IL5 accreditation — data stays within US Government cloud regions, fully ITAR compliant. HCL is an Azure Expert MSP (every year since launch) with all 21

		productivity for 185,000+ employees.	Microsoft specialisations and 76,000+ Microsoft-trained professionals.
Microsoft	**Copilot Studio + Azure AI Foundry**	Custom AI agent development, agentic orchestration, and enterprise copilot deployment for RTX business units. Power Platform for rapid AI workflow automation by citizen developers within RTX's CoE.	Deep SAP integration via Azure. ITAR-compatible deployment. HCL holds Microsoft Copilot Specialisation — RTX's agentic AI agents are built to Microsoft's enterprise security standards from Day 1.
AWS	**AWS GovCloud (US)**	Secondary cloud for Collins and Pratt & Whitney commercial aviation workloads, satellite data processing, and FlightAware analytics. AWS SageMaker / Bedrock for commercial model workloads.	RTX's December 2025 AWS Strategic Collaboration Agreement (100K+ EC2 instances, SageMaker/Bedrock, GovCloud) makes this a natural secondary cloud. HCL is AWS Premier Consulting Partner (highest tier) with 34,000+ trained resources.
Dell Technologies	**Dell EMC PowerEdge (Edge Servers)**	Edge compute at 500+ manufacturing facilities for real-time AI inference with sub-10ms latency. GTF engine test cell data processing. On-premises GPU inference for ITAR-classified quality AI models.	Purpose-built for manufacturing edge. Ruggedised for plant floor environments. NVIDIA GPU integration validated for computer vision QC — replicating RTX's Santa Isabel model: 14% output increase, inspection time cut from 30 to 10 minutes.

3.3 — OT/IT Convergence & Manufacturing

RTX operates 500+ manufacturing facilities with 250,000+ instrument tags currently disconnected from enterprise AI. This layer bridges that gap using existing RTX OEM relationships — Siemens (already running Teamcenter PLM at RTX), PTC (Windchill deployed), and Rockwell (present in Raytheon defense facilities) — minimising integration risk while maximising speed to first AI output at the factory floor.

OEM	Product / Module	Role in RTX EOS	Why Selected for RTX
Siemens	**Industrial Edge + MindSphere + Opcenter MES**	OT connectivity for P&W engine manufacturing, missile assembly, and Collins avionics plants. Connects CNC machining centres, engine test cells, and cleanroom sensors. Opcenter MES manages production execution.	RTX already runs Siemens Teamcenter PLM and Siemens MES — same OEM reduces integration risk and accelerates deployment. Siemens Industrial Edge certified for aerospace. Micron-level precision monitoring directly addresses the GTF blade quality crisis.
AVEVA (Schneider)	**AVEVA PI System (OSIsoft)**	Industrial data historian aggregating 250,000+ instrument tags across 500+ RTX facilities. Real-time streaming to AI models for predictive maintenance. Foundation of the L3 data lakehouse.	Industry standard for defense and aerospace historians. Real-time OT-to-AI streaming enables HCL AI Force to detect GTF engine anomalies up to 1,000 hours in advance — consistent with P&W's existing ML oil analysis capability.
Rockwell Automation	**FactoryTalk + Plex MES**	Manufacturing execution and OT integration for Raytheon defense facilities — missile assembly, weapons systems manufacturing, and Redstone Arsenal (50% capacity expansion funded in 2025).	Rockwell already present in RTX's OT landscape. FactoryTalk provides safety-critical process data for Tier 1 AI monitoring. CMMC-compliant architecture for Raytheon defense manufacturing. HCL iPEO integrates natively with FactoryTalk.

PTC	**ThingWorx IIoT + Vuforia AR**	Industrial IoT connecting legacy OT assets without re-engineering. AR maintenance guidance overlays for GTF MRO technicians — digital work instructions on physical engines. Native Windchill PLM integration.	PTC Windchill already deployed by RTX for PLM. Vuforia AR proven for aircraft engine maintenance — reduces MRO turnaround by 25–40%, directly addressing P&W's 250–300 day GTF inspection backlog.

3.4 — AI, ML & Analytics

The AI/ML layer is built around HCL AI Force as the primary intelligence platform, with NVIDIA, Databricks, and DataRobot providing compute, data, and governance infrastructure. Note: Palantir Foundry has been deliberately excluded — its June 2025 Accenture partnership makes it a competitor platform. HCL's AI Force GRC Module provides equivalent classified data governance natively, without third-party dependency.

OEM	Product / Module	Role in RTX EOS	Why Selected for RTX
NVIDIA	**AI Enterprise + DGX Systems (On-Premises)**	GPU compute for AI model training — computer vision for GTF blade quality inspection, predictive maintenance for MRO, NLP for defense contract analysis. On-premises DGX for ITAR-classified model training.	Only GPU platform meeting aerospace precision AI requirements at micron-level quality detection. DGX deployable on-premises — no classified IP leaves RTX facilities. HCL's NVIDIA Physical AI Innovation Lab (Santa Clara) provides direct access to NVIDIA's most advanced aerospace AI capabilities.
Databricks	**Delta Lake + MLflow**	Unified data lakehouse and MLOps platform. End-to-end model lifecycle management — training, versioning, deployment, monitoring, and drift detection. Handles both real-time OT streams and batch SAP transactional data.	Defense-proven at scale. MLflow provides the model registry required by RTX's Tier 1/2/3 governance framework. Delta Lake handles heterogeneous data: 250,000 OT tags, SAP records, Teamcenter PLM data, and Collins FlightAware analytics.
DataRobot	**MLOps + AI Governance**	Automated ML for citizen data scientists within RTX's CoE post-Month 18. Bias monitoring, explainability dashboards, and model performance tracking. Bridges HCL-led AI deployment to RTX self-sufficiency by Month 36.	Purpose-built for regulated industries. Explainability engine satisfies RTX's Tier 1/2 audit requirements and supports RTX's Responsible AI Corporate Policy (adopted board-level May 2024).

3.5 — Security, Compliance & ITAR Governance

Every security OEM is selected specifically to address RTX's documented compliance gaps — Varonis for automated ITAR data classification, SailPoint for US-person-only access control, CrowdStrike for nation-state threat defence, and Palo Alto for continuous CMMC monitoring ahead of the November 2026 Phase 2 deadline.

OEM	Product / Module	Role in RTX EOS	Why Selected for RTX
CrowdStrike	**Falcon Platform (XDR + Zero Trust)**	AI-powered threat detection across IT and OT. Nation-state cyber threat defence for RTX's defense IP. Zero-trust architecture enforcement across the classified/unclassified boundary.	Industry benchmark for defense contractor cybersecurity. Required for CMMC Level 2/3 compliance (DFARS effective November 2025). HCL's Falcon deployment experience accelerates RTX's November 2026 C3PAO certification target.

Palo Alto Networks	**Prisma Cloud + Cortex XSOAR**	Cloud security posture management for Azure Gov + AWS GovCloud. Automated CMMC and NIST SP 800-171 compliance monitoring. AI model security event incident response automation.	CMMC Level 2/3 compliance tooling built in. XSOAR automates incident response for AI model security events — critical given RTX's $200M ITAR penalty history and NDAA Section 1513 AI cybersecurity mandate (June 2026 deadline).
SailPoint	**IdentityNow (IAM)**	Identity and access management for 185,000+ employees and 60,000+ contractors. Role-based access control for classified/unclassified boundary management. Security clearance status integrated into access decisions.	ITAR access control enforcement — restricts all defense technical data to US persons only. Supports 50,000+ cleared personnel with automated access provisioning. HCL AI Force GRC integrates with SailPoint for unified identity-aware AI governance.
Varonis	**Data Security Platform**	Data classification and governance for ITAR-controlled technical data. Automated detection of unauthorised cross-border data transfer. GDPR right-to-erasure workflow for EU employee data.	Purpose-built for ITAR/EAR compliance. Automatically classifies and tracks Controlled Technical Data and CUI — satisfying NIST SP 800-171's 110 controls. Directly addresses the 750+ ITAR violations that triggered RTX's $200M August 2024 penalty.

3.6 — Innovation & Experience Layer

The experience layer converts AI outputs into decisions. PTC Vuforia and Microsoft HoloLens 2 bring AI guidance directly to the hands of P&W MRO technicians and Raytheon defense engineers. All OEMs in this layer are additive to RTX's existing investments — ServiceNow and Salesforce are already deployed; HCL adds intelligence on top, not alongside.

OEM	Product / Module	Role in RTX EOS	Why Selected for RTX
PTC	**Vuforia AR (Smart Glasses / Tablet)**	AR maintenance guidance for P&W GTF engine MRO technicians. Digital work instructions overlaid on physical engine components. Hands-free operation in shop floor environments.	Proven 40% reduction in MRO turnaround for complex engine maintenance — directly applicable to P&W's 250–300 day GTF inspection backlog. Native PTC Windchill integration keeps digital work instructions in sync with engineering changes.
Microsoft	**HoloLens 2**	AR/MR for design review, missile assembly guidance, and immersive training for Raytheon defense engineers. Digital twin visualisation for RTX's Connected Factory Initiative.	DoD-approved device — cleared for classified defense manufacturing. Integrates with Azure Digital Twins for RTX's Connected Factory digital twin layer. Microsoft's existing RTX Strategic Collaboration ensures licensing continuity.
ServiceNow	**Now Platform + AI Ops**	ITSM AI automation — 60% L1 ticket deflection for 500+ application landscape. AI-powered change management workflows for the application rationalisation programme. Predictive incident prevention.	Already deployed in RTX's IT landscape — HCL AI Force.Ops adds agentic intelligence on top of the existing ServiceNow investment without replacement. Native integration with SAP and Microsoft platforms.
Salesforce	**Einstein AI + Data Cloud**	Commercial aerospace CRM intelligence — connecting Collins and P&W airline customer engine health data to commercial account management. Cross-	Already deployed at RTX. Engine health data from Ascentia PHM (1,300 aircraft tails) flows through HCL AI Force.Data into Salesforce Data Cloud — enabling proactive MRO pricing before airlines request services.

		segment aftermarket opportunity identification.	

4 Phased Implementation Roadmap

The HCL EOS is delivered across 4 structured phases over 36 months. Each phase has defined entry/exit criteria, milestone gates, and RTX Steering Committee sign-off. No phase proceeds without formal RTX approval — and no HCL fee is paid without Steering Committee sign-off on delivered outcomes.

4.1 — Four-Phase Delivery Overview

The roadmap is designed to generate measurable ROI from Month 3 — through Quick Wins — while building the enterprise platform in parallel. RTX's primary concern is business continuity; every phase is designed for zero disruption to live defense contracts, production systems, or operational commitments.

Phase	Name	Duration	Primary Focus	Key Milestone / Exit Criteria
0	Foundation & Discovery	Months 1–3	Current state mapping, process mining, use case backlog (100+), architecture ADRs, tech debt assessment, Quick Wins deployment	**100+ AI use cases scored by business value; 3–5 Quick Wins live; architecture blueprint approved by RTX Steering Committee**
1	Core Platform & Priority AI	Months 4–12	EOS infrastructure, SAP BTP integration, OT/IT fabric, AVEVA PI connected, Databricks lakehouse live, 25 priority AI use cases in production	**25 AI use cases live; CoE Phase 1 operational; P&W GTF blade AI QC deployed; 60% L1 ticket deflection achieved**
2	Scale & Optimise	Months 13–24	Full domain AI deployment, SAP S/4HANA go-live, 25 additional AI use cases, tech debt reduction wave, CoE transition to 60/40 RTX-led	**50% applications rationalised; S/4HANA go-live; CoE 60% RTX-led; financial close at 6 days sustained**
3	Autonomy & Innovation	Months 25–36	Full agentic orchestration at scale, digital twin deployment, RTX AI self-sufficiency, innovation fund, CoE full handover	**CoE 80%+ RTX-owned; $250M+ IT budget freed; 60+ AI use cases in production; innovation fund operational**

4.2 — Phase 0: Quick Win Use Cases

Quick Wins are deployed in Months 2–3 to generate early ROI and Steering Committee confidence before the main platform build begins. Each Quick Win is selected for high business value, low technical complexity, and fast time-to-production — no Quick Win takes longer than 8 weeks from kickoff to live deployment.

Quick Win Use Case	Domain	OEM Stack	Target Outcome
AI-Powered L1 IT Ticket Deflection	IT Ops	ServiceNow AI Ops	**30–40% L1 deflection within 90 days**
Contract NLP Analysis Accelerator	Defense / Legal	Azure AI + SAP BTP	**60% reduction in contract review time**
Predictive Maintenance Pilot (P&W GTF)	Manufacturing	AVEVA PI + Databricks + AI Force	**First engine health alerts replacing manual inspection**

Supply Chain Risk Dashboard	Supply Chain	SAP Ariba + Power BI + iSCM	**Real-time risk visibility across top 500 at-risk suppliers**
Financial Close Automation Pilot	Finance	SAP S/4HANA + AI Core	**First automated journal entries and reconciliation**

4.3 — Phase 1: 25 Priority AI Use Cases

The 25 priority use cases across all 6 functional domains are sequenced by business value, data readiness, and RTX urgency. The GTF blade quality AI and missile assembly QC are deployed first — directly addressing RTX's $6–7B GTF crisis and Raytheon's $268B defense backlog conversion challenge.

- GTF Engine Blade Quality AI (P&W) — NVIDIA CV + AVEVA PI: 30% scrap reduction; micron-level defect detection
- Missile Assembly QC Verification (Raytheon) — NVIDIA CV + Siemens MES: 99.99% quality verification
- Predictive Maintenance, Engine Test Cells (P&W) — AVEVA PI + Azure ML: 35% unplanned downtime reduction
- Supply Chain Risk AI, 14,000 Suppliers — SAP Ariba + Databricks + iSCM: real-time risk scores, early warning
- Rare Material Sourcing Intelligence (Ti/Co) — SAP IBP + AI Force: geopolitical risk scoring, alternative sourcing
- ITAR Compliance Automation — Varonis + SAP Ariba + AI Force GRC: automated ITAR flag on all cross-border shipments
- Engine MRO Aftermarket Demand Prediction (P&W) — SAP IBP + Salesforce Einstein: 15% inventory optimisation
- Financial Close Acceleration — SAP S/4HANA + AI Core + Joule: 10 days to 6 days; automated reconciliation
- EVMS AI Reporting — SAP PS + AI Core: automated earned value management reporting for all defense programs
- CAS Compliance Monitoring — SAP FI + AI Force GRC: continuous compliance, eliminates manual audit preparation
- Security Clearance Pipeline Optimiser — Workday + Azure AI + SailPoint: reduce 8–24 month clearance bottleneck
- Engineering Talent Retention Prediction — Workday + DataRobot: identify flight-risk engineers 90 days ahead
- AI-Powered L1 Ticket Deflection (Full Scale) — ServiceNow AI Ops + AI Force.Ops: 60% deflection achieved
- Nation-State Threat Detection — CrowdStrike Falcon + Palo Alto Cortex: real-time defense IP protection
- Application Rationalisation Intelligence — ServiceNow + AI Force: AI-scored candidates across 500+ apps

4.4 — Phase 3: Tech Debt Resolution at Month 36

By Month 36, RTX transitions from a fragmented, legacy-dependent enterprise to a self-sufficient AI-powered organisation. The table below captures the full transformation — every metric is a contractual KPI tied to HCL's outcome-based fee structure.

Metric	Month 0 Baseline	Month 36 Target	Value Delivered
Total Applications	500+	**~120–180**	70%+ application footprint eliminated
IT Budget on Maintenance	62%	**38%**	$250M+ freed annually for innovation
AI Use Cases in Production	0 (disconnected POCs)	**60+**	Full enterprise AI across all 6 domains
Financial Close	10+ business days	**6 business days**	$15M+ annual efficiency gain
L1 Ticket Deflection	0%	**60%**	$8M+ annual IT ops savings
Unplanned Downtime	Baseline	**–35%**	$40M+ annual manufacturing savings
OT Connected Facilities	0	**500+**	250,000+ tags feeding enterprise AI
CoE Ownership	0% RTX-led	**80%+ RTX-led**	Full AI self-sufficiency — no HCL dependency

4.5 — 4-P AI Impact Framework & 100+ Use Case Backlog

> *To move RTX beyond pilot purgatory — where 72% of enterprises stall — every use case in this backlog is scored using the 4-P AI Impact Formula: AI Impact = People × Programmability × Productivity × Precision. This is not a list of IT projects. It is a Strategic Value Advisory that determines exactly where HCL deploys capital first, and why.*

The 4-P Scoring Engine

The formula produces a deterministic score for every use case, ensuring capital is allocated to the highest-yield Reshape and Invent plays — not the politically easiest or technically simplest. The table below shows how each P is calibrated to RTX's specific operational reality.

P Factor	What It Measures	RTX Calibration	Why It Matters for This Bid
People	Headcount in the target process	**RTX's 185,000+ employees and 60,000 contractors**	Higher headcount = greater leverage; manufacturing plant workers, finance teams, and cleared engineers each carry different cost weights
Programmability	% of workflow currently digitised	**~95% in IT/ERP; ~60% in high-precision OT plant floor**	Low programmability = high Phase 0 investment; high programmability = fast path to production
Productivity	Hours saved/week × specific labour cost	**$120/hr for P&W engineers; $42/ticket for IT L1**	Translates AI output directly into RTX's financial model — not generic benchmarks
Precision	Required accuracy threshold	**99.5%+ for Tier 1 Safety-Critical (GTF blade QC)**	Drives architecture choices: on-premises GPU, human-in-loop, air-gapped inference

D.R.I. Portfolio Classification

Every use case is classified into one of three play types before scoring. This prevents over-investment in safe Deploy plays at the expense of transformative Reshape and market-creating Invent plays — the pattern that traps most AI programmes in incremental improvement.

Play	Characteristics	# Use Cases	Role in the RTX EOS Engagement
Deploy	**Quick, high-confidence, lower complexity**	~30 use cases	**Foundation plays — fast to production (8–12 weeks), primarily extending existing RTX technology investments. Generates early ROI and stakeholder confidence for the Steering Committee by Month 3.**
Reshape	**Core transformation, highest total value**	~55 use cases	**The majority of the backlog — use cases that fundamentally change how RTX operates in a domain. Typically 12–16 week sprints via One-Pizza Agentic Pods. This is where the $2.2B return is built.**
Invent	**Novel AI, longer horizon, market-creating**	~15 use cases	**SmarTwin, Readiness-as-a-Service, SIPR-boundary agents — plays RTX's competitors are not running yet. Funded through the Innovation Fund (Section 6.1), governed quarterly, outcomes validated before renewal.**

Domain 1: Manufacturing & Plant Operations — Rank #1

The GTF engine crisis is RTX's most urgent AI opportunity. 835 grounded aircraft, 250–300 day inspection turnaround, and $6–7B in quality costs create a burning platform. HCL AI Force's computer vision module combined with iMRO/4 is the direct answer — deployed on-premises for ITAR compliance, targeting 99.5%+ precision threshold.

1	Manufacturing & Plant Operations		4-P Score: 5,670	Rank #1	
#	**Play**	**Use Case & Solution**	**4-P Impact Metric**	**HCL IP / Platform**	**Est. Value**
1	Reshape	GTF Vision Quality Pipeline: HCL AI Force computer vision + iMRO/4 for automated turbine blade inspection at micron-level. Deployed on NVIDIA DGX on-premises — ITAR-compliant, no cloud egress.	Precision: 99.5%+ threshold eliminates manual rework loops for GTF high-pressure turbine disks (contaminated powder metal, Oct 2015–Sept 2021 batch).	**HCL AI Force (CV module) + iMRO/4 + NVIDIA DGX on-prem**	**$1.5B+**
2	Reshape	Single-Micron Predictive Quality: Real-time variance detection in turbine disk milling using AVEVA PI sensor streams and Databricks ML. Alert before defect, not after.	Productivity: 35% reduction in unplanned downtime across P&W engine plants. ML oil analysis already detects issues 1,000 hours ahead — this extends it to machining.	**HCL AI Force.Data + AVEVA PI + Databricks + NVIDIA**	**$250M**
3	Deploy	Lighthouse Factory Scale-Out: Replicating the 14% output boost from Collins Santa Isabel (30-min inspection → 10 min, defect escapes halved) across all 500+ facilities.	People: Impacts manufacturing workforce across 30+ countries. Standard architecture — deploy once, replicate at scale using HCL's Base90 factory templates.	**HCL Base90 + AI Force + Siemens Industrial Edge + Dell EMC Edge**	**$400M**
4	Invent	Autonomous SmarTwin Maintenance: Full digital twin of P&W GTF engine manufacturing — predicts failures 1,000 hours in advance using physics simulation + AI quality models.	Programmability: Bridges the OT/IT gap at the engine test cell. AVEVA PI + HCL iPEO creates the real-time data layer the twin requires.	**HCL iPEO + AVEVA PI + Azure Digital Twins + NVIDIA Physical AI**	**$200M**
5–25	Reshape	Other: Missile assembly AI verification (Raytheon), hazardous area robotic inspection (Redstone Arsenal), additive manufacturing real-time layer monitoring, cleanroom contamination detection.	Various precision and productivity impacts across all three RTX business segments. Each use case scored individually in the full 100+ backlog.	**HCL AI Force + iMRO/4 + Siemens MES + Rockwell FactoryTalk**	**$300M+**

Domain 2: Supply Chain & Logistics — Rank #2

RTX's 14,000+ supplier network is a compounding risk — tariffs, geopolitical concentration, rare material dependencies, and ITAR automation gaps all compound. The Guardian Supplier Risk Engine and ITAR Logistics Automator address RTX's two most immediate supply chain vulnerabilities with precision-first AI.

2	Supply Chain & Logistics		4-P Score: 3,072	Rank #2	
#	**Play**	**Use Case & Solution**	**4-P Impact Metric**	**HCL IP / Platform**	**Est. Value**
26	Reshape	Guardian Supplier Risk Engine: AI monitoring 14,000+ suppliers for geopolitical shocks, single-	Precision: High-fidelity demand forecasting targeting 15% MAPE. Addresses RTX's	**HCL iSCM + SAP Ariba + AI**	**$850M**

		source concentration risk, and tariff exposure. Real-time risk scores surfaced in SAP Ariba.	$850M 2025 tariff exposure and rare material (Ti/Co/Re) sourcing bottlenecks.	**Force.Data + Azure AI**	
27	**Reshape**	ITAR Logistics Automator: Automated ITAR/EAR classification for all cross-border shipments via SAP GTS. AI Force GRC flags violations before they occur — not after a $200M penalty.	Precision: Eliminates repeat of RTX's 750+ ITAR violations. Automated cross-border data controls with AI Force GRC input/output scanning on every shipment record.	**HCL AI Force GRC + SAP GTS + Varonis + SailPoint**	**$200M+**
28	**Invent**	Readiness-as-a-Service: AI-driven aftermarket parts demand prediction connecting 1,300+ aircraft tail health data (Ascentia PHM) to MRO parts planning. Reduces product unavailability by 65%.	Productivity: 65% reduction in parts unavailability for airline customers. Engine health → demand signal → pre-positioned spares before airlines request MRO.	**HCL iSCM + SAP IBP + Salesforce Einstein + AI Force.Data**	**$350M**
29–45	**Deploy**	Other: Automated SAP Ariba procurement audits, rare material sourcing intelligence (geopolitical scoring), warehouse autonomous stock counting, demand signal integration from airline customers.	Various productivity and precision impacts. Full 4-P scores in the Phase 0 use case backlog delivered to RTX Steering Committee by end of Month 1.	**HCL iSCM + SAP Ariba/IBP + AI Force + Databricks**	**$200M**

Domain 3: Finance & Accounting — Rank #3

Finance has the highest Programmability score of any domain — 80%+ of RTX's financial processes already run on SAP S/4HANA. This means the path from use case to production is shorter here than anywhere else. The Autonomous Close Engine and AI-Augmented Contract Pricing are the fastest value generators in the backlog after Manufacturing.

3	**Finance & Accounting**		4-P Score: 3,024	**Rank #3**	
#	**Play**	**Use Case & Solution**	**4-P Impact Metric**	**HCL IP / Platform**	**Est. Value**
46	**Reshape**	Autonomous Close Engine: AI-powered reconciliation targeting the 6-day financial close. SAP S/4HANA auto-reconciliation rate moves from 15% to 80% — four fewer days of manual close cycles.	Programmability: High digitisation (80%+) in SAP S/4HANA core makes this the fastest path to production. HANASmart accelerates the ECC-to-S/4HANA migration enabling this use case.	**HCL AI Force for SAP + HANASmart + SAP S/4HANA + SAP Joule**	**$100M**
47	**Reshape**	AI-Augmented Contract Pricing: Predictive modelling for defense cost-plus and fixed-price bids. Accurate Estimate-at-Completion (EAC) updates across RTX's $107B defense backlog.	Precision: EAC accuracy directly determines profitability on long-cycle defense programs. AI Force GRC ensures classified contract data stays within ITAR boundaries during model inference.	**HCL AI Force GRC + SAP PS + Azure AI (Gov Cloud)**	**$500M+**
48	**Reshape**	Automated CAS/EVMS Compliance: AI-driven reporting for Cost Accounting Standards and Earned Value Management System. Auto-reconciliation removes the quarterly manual audit burden.	Productivity: Auto-reconciliation rates from 15% to 80%. Eliminates the finance team capacity consumed by manual CAS preparation — frees analysts for higher-value work.	**HCL AI Force for SAP + SAP PS + DataRobot Governance**	**$80M**

49–65	Deploy	Other: DSO improvement via automated AR reconciliation, travel and expense fraud detection, tax audit AI, FX exposure prediction, inter-company elimination automation.	Various productivity impacts across RTX's global finance function. Each use case pre-scored against RTX's SAP digitisation baseline in the Phase 0 assessment.	**HCL AI Force for SAP + SAP S/4HANA + Azure AI**	**$120M**

Domain 4: Defense Programs — Rank #4

Defense Programs is the domain most underserved by current AI deployments at RTX — and the highest-risk gap. Raytheon's $107B defense backlog and $268B total backlog conversion depend on accurate EAC prediction, competitive bid pricing, and eventually, AI operating across the classified boundary. HCL AI Force GRC's on-premises and air-gapped deployment modes make this possible without creating new ITAR exposure.

4	Defense Programs (Raytheon)		4-P Score: 2,800	Rank #4	
#	**Play**	**Use Case & Solution**	**4-P Impact Metric**	**HCL IP / Platform**	**Est. Value**
66	Reshape	Defense Contract Win Rate AI: AI-augmented bid pricing for cost-plus and FFP contracts. Predictive win/loss modelling trained on RTX's historical bid data within ITAR-compliant on-premises inference.	Precision: Improved bid accuracy on Raytheon's $107B defense backlog. AI Force GRC on-premises deployment — no classified contract data leaves RTX facilities.	**HCL AI Force GRC (on-prem) + SAP SD + Azure Gov AI**	**$500M+**
67	Reshape	Program EAC Intelligence: AI-driven Estimate-at-Completion prediction for 10–20 year defense programs. Flags cost overrun risk 6–12 months earlier than traditional EVMS methods.	Precision: Earlier EAC warning = smaller corrective action. Directly addresses Raytheon's need to convert the $268B backlog without margin erosion on complex long-cycle programs.	**HCL AI Force for SAP + SAP PS + DataRobot**	**$250M**
68	Invent	SIPR-Boundary Agentic AI: AI agents operating across the classified/unclassified boundary with hard enforcement via AI Force GRC. Enables classified program management AI without human data-handling overhead.	Programmability: Eliminates the classified data handling bottleneck — the single biggest barrier to AI adoption in Raytheon's defense programs division.	**HCL AI Force GRC (air-gapped) + Azure Government (IL5)**	**$300M**
69–72	Deploy	Other: Government contract compliance monitoring (DFARS automated audit), classified document summarisation (on-prem LLM), bid team productivity AI, RCADE-adjacent simulation AI.	Various precision and productivity impacts. All use cases deployed within FedRAMP High / air-gapped environments with full ITAR compliance from Day 1.	**HCL AI Force GRC + AI Force.Ops + Azure Gov**	**$150M**

Domain 5: IT Operations & Cybersecurity — Rank #5

IT Operations is the domain with the highest Programmability and the most immediate cost lever — 62% of RTX's IT budget consumed by legacy maintenance. The Sovereign Helpdesk Agent and SAP Migration Accelerator attack this directly. Every use case in this domain is additive to RTX's existing ServiceNow and SAP investments — no rip-and-replace.

5	IT Operations & Cybersecurity	4-P Score: 2,646	Rank #5

#	Play	Use Case & Solution	4-P Impact Metric	HCL IP / Platform	Est. Value
73	Reshape	Sovereign Helpdesk Agent: HCL AI Force.Ops for 60% L1 deflection across 50,000+ monthly tickets in ITAR air-gapped environments. Cost per resolved ticket drops from $42 to under $2.	Productivity: $40M+ annual IT ops savings. ServiceNow AI Ops integration means zero rip-and-replace — intelligence adds on top of RTX's existing ITSM investment.	**HCL AI Force.Ops + ServiceNow AI Ops + AI Force GRC**	**$150M**
74	Reshape	OneRTX SAP Migration Accelerator: HANASmart + GenAI automating custom ABAP code migration from ECC to S/4HANA. Converts months of manual re-coding into days of automated migration.	Programmability: 95% digitisation in IT-led ERP consolidation. Directly de-risks the OneRTX PST build phase ahead of SAP ECC end-of-support December 2027.	**HCL HANASmart + AI Force.Software + SAP BTP**	**$250M**
75	Deploy	App Rationalisation AI: Automated assessment of 500+ applications — AI scores each against usage data, integration complexity, and business value to target 70% retirement by Month 36.	Productivity: IT maintenance budget from 62% to 38% — freeing $250M+ annually for innovation. AI Force.Ops provides the usage telemetry layer without additional instrumentation.	**HCL AI Force.Ops + ServiceNow + Azure AI**	**$250M+**
76–85	Reshape	Other: Classified threat hunter (CrowdStrike + AI Force GRC), automated test case generation (on-prem LLM), cloud cost optimisation AI, zero-trust posture scoring.	Various IT productivity and security precision impacts. All use cases on the Phase 1 IT Ops track are live within 12 weeks of platform deployment.	**HCL AI Force.Ops + CrowdStrike + AI Force.Software**	**$180M**

Domain 6: HR & Safety — Rank #6

The security clearance bottleneck (8–24 months) is RTX's hidden workforce constraint — it limits how fast Raytheon can ramp for the $268B defense backlog. The Clearance Pipeline Optimiser does not change government processes; it maximises RTX's control over the variables it can influence: document completeness, submission timing, and processing status prediction.

6	HR & Safety	4-P Score: 1,250	Rank #6		
#	**Play**	**Use Case & Solution**	**4-P Impact Metric**	**HCL IP / Platform**	**Est. Value**
86	Reshape	Clearance Pipeline Optimiser: AI-driven tracking and intelligent triage of 50,000+ active security clearances. Predicts processing bottlenecks and recommends pre-submission document completion to cut 8–24 month timelines.	People: Affects 185,000 employees and every defense program ramp-up. Security clearance speed is RTX's #1 workforce constraint — AI reduces the bottleneck without changing government processes.	**HCL AI Force.Data + Workday + SailPoint + Azure AI**	**$200M**
87	Reshape	Retention Early Warning: Predictive modelling for engineering talent attrition targeting less than 5% annualised loss. Identifies flight-risk engineers 90 days before resignation across RTX's 7.1% aerospace attrition baseline.	Productivity: Prevents the cost of replacing highly specialised defense engineers — cleared aerospace engineers take 2–3 years and $150K–$300K to replace. One prevented attrition event pays for months of AI platform cost.	**HCL AI Force.Data + Workday + DataRobot + Azure AI**	**$100M**

88–100	**Deploy**	Other: ITAR compliance auditor for technical data exports (automated US-person access verification), operator fatigue monitoring via wearables at Raytheon defense facilities, post-merger cultural integration analytics for Collins/P&W/Raytheon segment alignment.	Various People and Precision impacts. Wearable fatigue monitoring is a Tier 1 safety use case — requires full explainability and human override governance per the AI Risk Framework.	**HCL AI Force + AI Force GRC + Workday + Azure AI**	**$150M**

Total Backlog Value Summary

The 100+ use cases across all 6 domains represent a total addressable value of $7.5B+ over the 5-year engagement period. The $2.2B baseline ROI in the Commercial Model (Section 6) uses conservative 60th-percentile capture rates — the full backlog represents the ceiling if RTX and HCL execute at pace across all domains.

Domain	4-P Rank & Score	Est. Total Value	Key Use Cases
Manufacturing & Plant Ops	Rank #1 | 4-P: 5,670	**$2.65B+**	GTF crisis ($1.5B) + lighthouse scale-out ($400M) + SmarTwin ($200M) + missile assembly + other (25 use cases total)
Supply Chain & Logistics	Rank #2 | 4-P: 3,072	**$1.6B+**	Guardian supplier risk ($850M) + Readiness-as-a-Service ($350M) + ITAR automator ($200M) + other (20 use cases)
Finance & Accounting	Rank #3 | 4-P: 3,024	**$800M+**	AI contract pricing ($500M) + autonomous close ($100M) + CAS/EVMS ($80M) + other (20 use cases)
Defense Programs	Rank #4 | 4-P: 2,800	**$1.2B+**	Win rate AI ($500M) + SIPR agentic AI ($300M) + EAC intelligence ($250M) + other (7 use cases)
IT Operations & Cyber	Rank #5 | 4-P: 2,646	**$830M+**	Contract pricing AI ($250M) + SAP migration ($250M) + app rationalisation ($250M) + other (13 use cases)
HR & Safety	Rank #6 | 4-P: 1,250	**$450M+**	Clearance optimiser ($200M) + retention AI ($100M) + ITAR compliance auditor + other (15 use cases)

Operationalising the Backlog: One-Pizza Agentic Pods

HCL does not plan multi-year mega-projects. Each use case is delivered by a dedicated One-Pizza Agentic Pod — a cross-functional team of 6–8 members (architect, AI engineer, domain SME, RTX co-secondee, QA, and change lead) operating in 12–16 week rapid-build sprints. This model has delivered a 13x increase in human capacity at comparable enterprises as workforces transition from manual execution to orchestrating autonomous AI systems. Every Pod follows the IMPACT Delivery Lifecycle: Identify → Model → Pilot → Accelerate → Certify → Transform. No use case exits Pilot without RTX Steering Committee sign-off. No use case enters production without passing the AI Risk Tier governance gate.

5 Governance, Risk & Compliance Framework

RTX's $200M ITAR penalty (August 2024) and board-level AI governance mandate (May 2024) mean compliance is not a feature — it is a precondition. HCL's AI Force GRC Module, 7 ITAR-compliant US delivery centres, and Butler Aerospace clearances make HCL the only partner who can deploy AI at RTX without creating new compliance exposure.

5.1 — AI Risk Tier Classification

Every AI use case in the RTX EOS is classified into one of three risk tiers before deployment. Tier assignment is determined jointly by HCL and RTX's AI governance committee — it is not unilaterally decided by HCL. No Tier 1 AI action executes without explicit, time-stamped human authorisation; this is hard-coded in AI Force Agent Studio, not a configuration parameter.

Tier	Category	RTX Examples	Governance Requirement
Tier 1	**Safety-Critical**	GTF blade QC AI; missile assembly verification; engine test cell process control; cleanroom contamination monitoring; additive manufacturing real-time control	**Full explainability + hard human approval gate + kill-switch + real-time monitoring + tested rollback plan. No Tier 1 AI action executes without explicit human authorisation.**
Tier 2	**Financial / Regulatory**	EVMS reporting AI; CAS compliance monitoring; defense contract pricing; SOX-relevant financial close automation; ITAR classification decisions	**Regulatory-grade audit trail + SOX-compatible logging + CFO/Controller approval gate. All Tier 2 outputs are immutably logged and reviewed quarterly.**
Tier 3	**Productivity / Advisory**	L1 ticket deflection; engineering talent analytics; supply chain opportunity identification; board-level dashboards; HR workforce planning	**Basic logging + user feedback loop + quarterly model accuracy review. Tier 3 actions are advisory only — no automated process changes without human initiation.**

5.2 — Regulatory Compliance Matrix

HCL's technical controls are designed to address RTX's specific regulatory obligations — not generic enterprise AI governance. The matrix below maps each regulation to the specific HCL and OEM controls that enforce it, with an RTX-specific rationale for why each control was selected.

Regulation	Scope	HCL Technical Controls	RTX-Specific Rationale
ITAR	All defense technical data, export controls	Varonis automated ITAR data classification + SailPoint US-person-only access control + Azure Gov US-only data residency + AI Force GRC input/output scanning	**HCL's 7 ITAR-compliant US delivery centres + Butler Aerospace security clearances ensure zero deemed-export risk on HCL's side.**
CMMC Level 2/3	All defense contract data (DFARS effective Nov 2025)	CrowdStrike Falcon + Palo Alto Prisma Cloud + NIST 800-171 controls mapped to SAP BTP security policies + continuous compliance scanning	**C3PAO third-party certification required by November 2026. HCL's deployment timeline puts RTX on track with 3+ months of buffer.**

DFARS	Defense acquisition compliance	AI Force GRC audit trail + SAP compliance documentation automation + DataRobot explainability engine for all AI-assisted decisions	**NDAA Section 1513 (FY2026) mandates AI/ML cybersecurity framework — AI Force GRC provides the framework, June 2026 Congressional deadline met.**
SOX	NYSE-listed financial reporting	SAP S/4HANA audit log + AI Core model registry + DataRobot Tier 2 governance + CFO dashboard with immutable decision trail	**Every AI-assisted financial close action has a provable, auditable chain from data input to output — CFO-ready from Day 1.**
CAS	Cost accounting for defense contracts	SAP PS + AI Force for SAP automated CAS compliance monitoring + continuous audit trail for all cost-plus and FFP contracts	**Automated CAS monitoring eliminates the manual audit preparation burden currently consuming significant finance team capacity.**
AS9100 / NADCAP	Aerospace quality management	Siemens MES + NVIDIA CV quality AI + digital work instructions with embedded AS9100 checklists + iMRO/4 quality process documentation	**AS9100-compliant quality AI for every manufacturing process — from GTF blade machining to missile sub-component assembly.**
NIST SP 800-171	Controlled Unclassified Information (CUI)	CrowdStrike XDR + Palo Alto XSOAR + SailPoint IAM + AI Force GRC continuous compliance scanning across all 110 controls	**All 110 NIST SP 800-171 controls mapped to the HCL EOS architecture — compliance is built in, not bolted on.**

6 Commercial Model & ROI

HCL accepts outcome-based commercial terms because we are confident in our delivery. Our fee is tied to RTX's results, not our effort. The $268B backlog, $10B+ annual technology investment, and board-level AI mandate mean RTX has both the need and the budget — and HCL has the platform, the credentials, and the skin in the game to deliver.

6.1 — Pricing Structure

The commercial model is structured to align HCL's incentives with RTX's outcomes at every stage. Platform licences pass through at market rates with full switch rights — RTX is never locked in. HCL's delivery fees are milestone-gated, and up to 20% of contract value is at risk against KPI performance.

Revenue Stream	Commercial Model	Description
Platform & Infrastructure	**Transparent licensing pass-through**	SAP, Azure Gov, NVIDIA, Databricks, CrowdStrike — all at published market rates with annual review and full switch rights. RTX retains all platform licences at engagement end.
HCL Delivery Services	**Fixed price per phase with milestone payments**	Phase gates trigger payments — no payment without RTX Steering Committee sign-off on delivered outcomes. HCL bears full cost of any phase that fails its exit criteria.
Outcome-Based Fee	**KPI-linked incentive (up to 20% of contract value)**	Tied to: IT budget reduction %, financial close days, manufacturing downtime %, L1 deflection %, AI use cases in production. HCL's upside is RTX's upside.
Managed Services (AMS)	**SLA-backed monthly recurring fee**	24/7 support with financial SLA penalties for breach. Continuous improvement built into the managed service model — HCL bears the cost of any regression against baseline KPIs.
Innovation Fund	**Quarterly governed innovation budget**	Jointly governed robotics, AR/VR, quantum, and responsible AI experimentation. Outcomes validated against business case before each quarterly renewal — no automatic spend.

6.2 — 5-Year ROI Model

The ROI model is built on RTX's actual reported financials — $88.6B revenue, ~$2B estimated IT budget, $6–7B GTF crisis costs — not generic industry benchmarks. Baseline figures are deliberately conservative; optimistic figures reflect what comparable AI deployments have achieved at similar-scale A&D enterprises.

Value Driver	Baseline /yr	Optimistic /yr	Calculation Basis	5-Year Cumulative
IT Budget Freed (maintenance to innovation)	$250M/yr	$350M/yr	62% to 38% of ~$2B IT budget	**$1.25B+**
Manufacturing Downtime Reduction (35%)	$40M/yr	$60M/yr	P&W + Collins + Raytheon facilities	**$200M+**

Financial Close Acceleration (4 days)	$15M/yr	$20M/yr	4-day reduction × finance team cost	**$75M+**
L1 IT Ticket Deflection (60%)	$8M/yr	$12M/yr	50,000 tickets/month × cost per ticket	**$40M+**
Supply Chain Risk Avoidance	$30M/yr	$50M/yr	Sole-source disruption prevention	**$150M+**
Engine Blade Scrap Reduction (30%)	$20M/yr	$35M/yr	P&W GTF blade scrap cost reduction	**$100M+**
Defense Contract Win Rate Improvement	$50M/yr	$80M/yr	AI-augmented pricing on bid pipeline	**$250M+**
MRO Aftermarket Pricing Optimisation	$25M/yr	$40M/yr	Engine health data-driven pricing	**$125M+**

6.3 — Scenario Analysis

The baseline scenario ($2.2B, 18-month payback) is used for all commercial structuring. Even in the pessimistic scenario — delivering only 60% of modelled benefits — the 28-month payback and $850M net value represent a compelling return on a $450M investment.

Scenario	5-Yr Benefit	5-Yr Investment	Net Value	Payback Period
Pessimistic (60%)	$1.3B	$450M	$850M	**~28 months**
Baseline (100%)	$2.2B	$450M	**$1.75B**	**~18 months**
Optimistic (140%)	$3.1B	$450M	$2.65B	**~12 months**

7 Why HCL Technologies

HCL is the only firm combining ITAR-cleared US delivery centres + patented AI platform (AI Force) + SAP GSSP with A&D-specific templates (Base90 / iMRO/4) + #1 aerospace & defense engineering heritage by Zinnov — rated Leader in all six Gartner Magic Quadrant reports for global IT services, a distinction no competitor holds.

7.1 — HCL's Eight Differentiators for RTX

Each differentiator below is substantiated by a verifiable credential, a named client reference, or a specific RTX context. This is not positioning — it is evidence.

Patented AI Platform — AI Force	HCL's AI Force is not a wrapper around OpenAI or a rebranded cloud service — it is a patented, production-proven GenAI + Agentic AI platform live with 25+ enterprise clients. Model-agnostic, on-premises capable, and ITAR-deployable from Day 1. No competitor in this bid has equivalent proprietary AI IP.
SAP Global Strategic Services Partner (GSSP)	HCL holds the highest SAP partnership tier globally — GSSP — with 15,000–20,000 SAP consultants and an estimated $2.5–3B in SAP revenue. HCL was the first GSSP to go live on RISE with SAP for its own transformation, and co-created an AI & Cloud Native Lab with SAP in Munich in December 2024. For OneRTX PST, HCL's Base90 + HANASmart + AI Force for SAP is unmatched.
#1 in A&D Engineering by Zinnov	HCL is rated #1 in Aerospace and Defense Engineering by Zinnov — not #3, not a Leader, #1. With $2.1–2.2B in Engineering & R&D Services revenue and preferred partner status with 100+ of the top 250 global R&D spenders, HCL understands GTF engine manufacturing tolerances, ITAR compliance, EVMS, and CAS. We do not need to be educated.
ITAR-Compliant by Design — Not Retrofit	7 ITAR-compliant US delivery centres through subsidiary Butler Aerospace & Defense, with facility security clearances and AS9100 / CMMI Level 5 certifications. AI Force deploys on-premises with zero data egress for classified model weights. This is the architecture RTX needs — not a promise to build it later.
Proven A&D IP — iMRO/4, Base90, iSCM, iPEO	iMRO/4 was selected by HAESL in February 2026 and deployed at Jet Aviation. Base90 delivers 90% of S/4HANA A&D requirements out-of-box. These are live deployments at comparable aerospace organisations — not case studies from adjacent industries.
No Lock-In Architecture	Every OEM in the HCL stack is replaceable at annual review. All platform licences are held by RTX, not HCL. All data, models, and IP created during the engagement belong to RTX. HCL architectures for RTX's freedom — not HCL's dependency.
CoE-First Delivery — RTX Self-Sufficiency by Month 36	HCL's primary success metric is RTX's independence. By Month 36, the CoE is 80%+ RTX-led, 60+ AI use cases are in production, and RTX has no dependency on HCL for AI operations. We build to be replaced — that is our differentiator and the commercial model reflects it: HCL's fee is tied to RTX's outcomes, not HCL's continued presence.
Outcome-Based Confidence	HCL accepts up to 20% of contract value at risk against KPI performance. We accept this because we have delivered comparable transformations: one client received $700M+ in R&D cost reduction — 7x their initial objective. RTX's $268B backlog, $10B+ annual technology investment, and board-level AI mandate give us high confidence in these numbers.

7.2 — Competitive Differentiation

The table below positions HCL against the three most likely competitors for this bid — Accenture, IBM, and TCS/Infosys — across the dimensions that matter most for RTX's specific requirements.

Dimension	HCL Technologies	Accenture	IBM Consulting	TCS / Infosys
Proprietary AI Platform	**AI Force (patented, 25+ clients)**	Applied Intelligence (partner-dependent)	WatsonX (mature but slow)	WisdomNext / Topaz (limited proven scale)
SAP Partnership Tier	**GSSP (highest)**	GSI Partner	GSI Partner	GSI Partner
ITAR-Compliant US Centres	**7 (Butler Aerospace, cleared)**	Multiple (but no dedicated A&D subsidiary)	Limited	None for Indian-national delivery staff
A&D Engineering Heritage	**#1 by Zinnov, 30+ years**	Strong advisory, weaker engineering	Limited A&D manufacturing depth	Limited ITAR infrastructure
A&D-Specific IP (live)	**iMRO/4, Base90, iPEO, iSCM**	None proprietary	None in A&D	None proprietary
Outcome-Based Model	**Yes — up to 20% at risk**	Limited	Limited	Rare